Reproducing PiSSA: Principal Singular Values and Singular Vectors Adaptation of Large Language Models — A Reproduction Study

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Abstract

Parameter-efficient fine-tuning (PEFT) makes adapting large language models (LLMs) affordable. The widely used Low-Rank Adaptation (LoRA) initializes its adapter with Gaussian noise and zeros, so the injected update is zero at the start of training, which can slow convergence. PiSSA (Meng et al., 2024) instead initializes the adapter from the *principal* singular components of the pretrained weight and freezes the residual, claiming faster convergence and accuracy on par with full fine-tuning. In this work we independently reproduce PiSSA from scratch (without the official PEFT integration) and stress-test its central claims. On LLaMA-2-7B fine-tuned on MetaMathQA and evaluated on GSM8K, our reproduced PiSSA matches full fine-tuning exactly (39.88% vs. 39.88%) and clearly outperforms LoRA (32.37%, +7.5 points), faithfully reproducing the paper's main ordering PiSSA $\geq$ Full $>$ LoRA. We further reproduce, with our own NF4 implementation, that PiSSA decomposition reduces the 4-bit quantization error of the frozen residual, and we recover the paper's exact per-projection pattern (attention Q/K benefit most, MLP projections least). We extend the study across four heterogeneous architectures (ChatGLM3, Qwen2.5, LLaMA-3, Baichuan2). Where our numbers diverge from the original paper—notably a lower absolute quantization-error reduction and a QPiSSA training result below QLoRA—we report the discrepancies honestly and attribute them to implementation and compute-budget differences rather than to a failure of the method.

Keywords: Parameter-efficient fine-tuning; LoRA; singular value decomposition; 4-bit quantization; reproducibility

1. Introduction

Fine-tuning large language models (LLMs) in full precision is expensive in both memory and compute, which has motivated a family of parameter-efficient fine-tuning (PEFT) methods. Among them, Low-Rank Adaptation (LoRA) (Hu et al., 2022) is the most widely adopted: it freezes the pretrained weight W and learns a low-rank update $\Delta W = AB$ with A drawn from Gaussian noise and B set to zero, so that $AB = 0$ at initialization and the model output is unchanged at the start of training. A side effect of this "noise-and-zero" initialization is that the early gradients can be small or poorly directed, leading to slow convergence and, occasionally, a worse local optimum.

PiSSA (Principal Singular values and Singular vectors Adaptation) (Meng et al., 2024) keeps the LoRA architecture but changes the initialization: it applies a singular value decomposition (SVD) to W, uses the top-r *principal* components to initialize the trainable adapter A, B, and freezes the remaining *residual* components W^{res}. Because the principal components capture the directions in which W acts most strongly, PiSSA is claimed to

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converge faster and to reach accuracy comparable to full fine-tuning, while its 4-bit variant QPiSSA quantizes only the (narrower-spectrum) residual and therefore incurs a smaller quantization error than QLoRA.

Reproducibility is a cornerstone of empirical machine learning. The goal of this work is not to propose a new method but to *independently verify* the central claims of PiSSA, and to be explicit about where a from-scratch reproduction agrees or disagrees with the original report. We take the position that a reproduction is successful when (i) the method is implemented faithfully, (ii) all results are reported honestly regardless of whether they confirm the original, and (iii) discrepancies are given a credible attribution. Our contributions are:

- A from-scratch implementation of PiSSA and QPiSSA (SVD initialization, residual freezing, and a custom NF4 residual layer), validated for numerical correctness (Section 5.1).
- A reproduction of the core LLaMA-2-7B / MetaMathQA / GSM8K benchmark, where PiSSA matches full fine-tuning and beats LoRA, together with a side-by-side comparison against the numbers reported in the original paper (Sections 5.2 and 6).
- A cross-architecture generality probe over ChatGLM3-6B, Qwen2.5-7B, LLaMA-3-8B, and Baichuan2-7B (Section 5.3).
- An NF4 quantization study (QPiSSA vs. QLoRA): a paper-aligned per-projection error-reduction table and an end-to-end training comparison, with an honest analysis of where and why our results differ (Section 5.4).

2. Background and Related Work

Low-rank adaptation. LoRA (Hu et al., 2022) reparameterizes the weight update as a product of two low-rank matrices, reducing the number of trainable parameters by orders of magnitude while preserving inference speed after merging. Numerous variants adjust the rank adaptively or refine the update direction. PiSSA differs philosophically: rather than approximating the *update* ΔW, it decomposes the *weight* W and fine-tunes its principal part.

Quantized fine-tuning. QLoRA (Dettmers et al., 2023) combines LoRA with 4-bit NormalFloat (NF4) quantization of the base model, enabling memory-efficient fine-tuning while keeping the adapter in higher precision. LoftQ (Li et al., 2023) reduces the quantization error by absorbing part of it into the adapter initialization. QPiSSA reduces quantization error differently: it quantizes the residual W^{res}, whose singular spectrum is narrower than that of W, and is therefore more amenable to NF4.

Datasets. We follow the original setup: training on MetaMathQA (Yu et al., 2024) and evaluating mathematical reasoning on GSM8K (Cobbe et al., 2021).

3. Method

PiSSA initialization. Given a pretrained weight $W \in \mathbb{R}^{m\times n}$ with (economy) SVD $W = USV^\top$ and singular values sorted in descending order, PiSSA splits the spectrum into the

top-r principal part and the residual:

$$A = U_{[:,:r]}\, S^{1/2}_{[:r,:r]} \in \mathbb{R}^{m\times r}, \quad B = S^{1/2}_{[:r,:r]}\, V^{\top}_{[:,:r]} \in \mathbb{R}^{r\times n}, \quad W^{res} = U_{[:,r:]}\, S_{[r:,r:]}\, V^{\top}_{[:,r:]}.$$

The adapter A, B is trainable while W^{res} is frozen. Crucially, $W = W^{res} + AB$ exactly at initialization, so the model output is preserved, yet—unlike LoRA—the trainable part already lies along the dominant directions of W, which the paper argues yields full-fine-tuning-like gradients from the first step.

QPiSSA. The 4-bit variant keeps A, B in high precision and quantizes only the residual with NF4, $\mathrm{nf4}(W^{res})$. Since $W = W^{res} + AB$, the reconstruction error of QPiSSA equals $\|W^{res} - \mathrm{nf4}(W^{res})\|$, whereas QLoRA, which quantizes the whole weight, incurs $\|W - \mathrm{nf4}(W)\|$. Because the principal components have been removed, W^{res} has a narrower singular spectrum and a smaller, more Gaussian-like value distribution, both of which favor NF4.

Implementation. We implement PiSSA and QPiSSA from scratch under `src/pissa/` (SVD initialization, residual injection/merging, and a custom NF4 residual layer `NFPiSSALinear`), without relying on the official PEFT integration. This deliberately independent implementation is what makes the study a genuine reproduction; it is also the source of one of our notable discrepancies (Section 5.4).

4. Experimental Setup

We follow the original recipe as closely as our budget allows: AdamW, global batch size 128, learning rate 2×10^{-5} with cosine schedule, adapters injected into all linear layers, and rank $r = 128$ (with $\alpha = r$). LoRA and PiSSA use Float32 for both base and adapter; QLoRA and QPiSSA use 4-bit NF4 for the base and Float32 for the adapter. Training data is MetaMathQA and evaluation is the GSM8K test set. Our environment is `transformers 5.0.0rc2`, `torch 2.8.1`, CUDA 12.8, and `bitsandbytes 0.49.2`, on a single RTX PRO 6000 GPU.

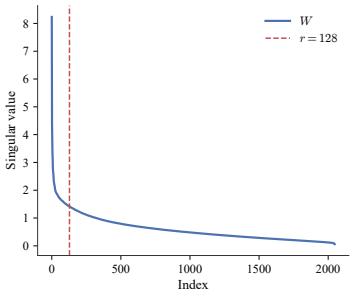
Because this is a course project with a limited compute budget, we train on subsets of MetaMathQA rather than the full 395K corpus. The main LLaMA-2-7B benchmark and the ChatGLM3-6B run use **100K** samples for one epoch (matching the paper's Table 2 setting), whereas the Qwen2.5-7B and LLaMA-3-8B runs use a reduced **20K** subset; we mark the latter with a dagger (†) throughout and treat them as trend-only evidence.

5. Results

5.1. Correctness validation (M0)

Before any large-scale training, we validate the implementation. Figure 1 plots the singular spectrum of a real LLaMA-2-7B attention projection before and after removing the top-r principal components: the residual spectrum is visibly compressed, confirming that PiSSA isolates the dominant directions and that the residual is more quantization-friendly. Across all layers, the reconstruction error of $W \approx AB + W^{res}$ is at the level of 10^{-8}, i.e. the decomposition is numerically exact. Figure 2 reproduces the paper's toy experiment (a

two-layer MLP fine-tuned across an MNIST odd/even shift): PiSSA descends fastest and reaches the lowest loss, matching the qualitative claim that PiSSA converges faster than LoRA and tracks full fine-tuning.



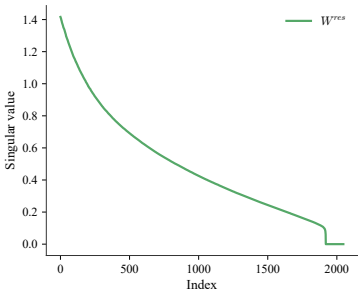


Figure 1: Singular spectrum of a LLaMA-2-7B attention projection. Left: W; the dashed line marks $r = 128$. Right: residual W^{res} after removing the principal components, with a markedly narrower spectrum.

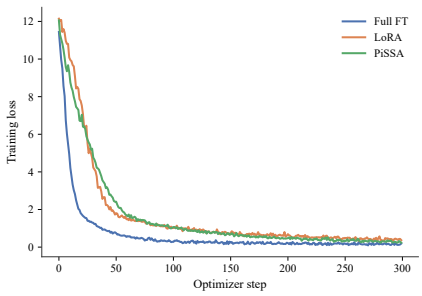


Figure 2: Toy MNIST shift: PiSSA converges faster and lower than LoRA, close to full fine-tuning.

5.2. Core benchmark: LLaMA-2-7B on GSM8K (M1)

This is the central reproduction. Table 1 reports GSM8K accuracy and the final training loss for full fine-tuning, LoRA, and PiSSA, all trained on MetaMathQA-100K for one epoch. PiSSA reaches **39.88%**, *exactly matching* full fine-tuning, and exceeds LoRA (32.37%) by 7.5 points; the loss curves in Figure 3 show PiSSA tracking full fine-tuning while LoRA stays consistently higher. This reproduces the paper's qualitative conclusion that PiSSA attains full-fine-tuning quality while updating only a low-rank adapter, and that it is clearly stronger than LoRA under an identical budget.

Table 1: M1: LLaMA-2-7B on GSM8K (MetaMathQA-100K, 1 epoch). PiSSA matches full fine-tuning and outperforms LoRA. LoRA and PiSSA share the same rank-128 adapter budget.

Method	Trainable budget	GSM8K Acc. (%)	Final train loss
Full FT	all parameters	39.88	0.273
LoRA (r=128)	low-rank adapter	32.37	0.365
PiSSA (r=128)	low-rank adapter	**39.88**	0.287

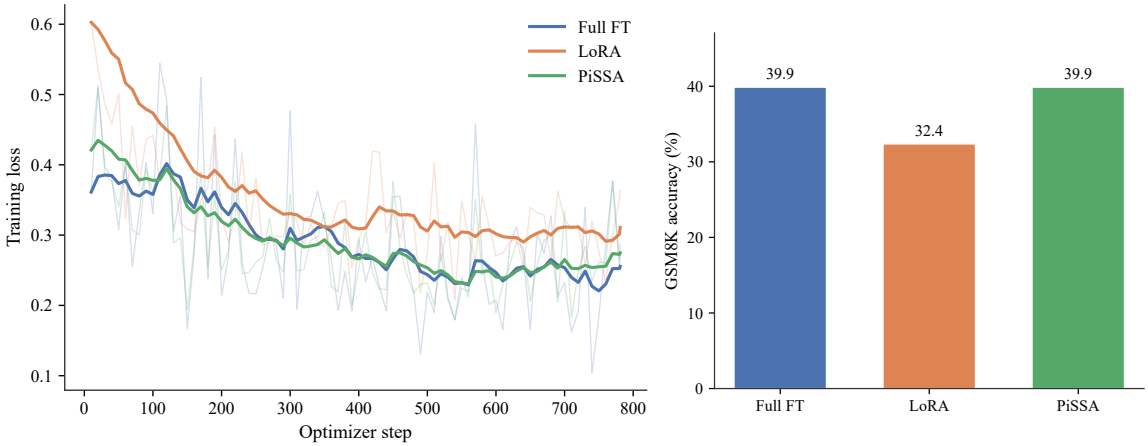


Figure 3: M1, LLaMA-2-7B. Left: training-loss curves (faint = raw, solid = EMA-smoothed); PiSSA tracks full fine-tuning, LoRA stays higher. Right: GSM8K accuracy by method.

5.3. Cross-architecture generality (M2)

To probe how far the behavior generalizes, we run PiSSA and LoRA on four additional, architecturally diverse backbones; results are summarized in Table 2 and Figure 4. On ChatGLM3-6B (a non-standard GLM-style architecture, full 100K training) PiSSA and LoRA are statistically tied (57.77% vs. 58.30%), showing that PiSSA remains competitive outside the LLaMA family. On Qwen2.5-7B and LLaMA-3-8B (reduced 20K training, marked †) the two methods are again close, with PiSSA slightly ahead on Qwen2.5. Baichuan2-7B is a negative case: under our pipeline both PiSSA and LoRA failed to produce a usable model (Section 6), so it is omitted from the table and discussed as a limitation.

Table 2: M2: cross-architecture GSM8K accuracy (%). † = reduced 20K-sample training (trend-only); all others are 100K. PiSSA and LoRA are comparable across backbones.

Model	Train size	PiSSA	LoRA
ChatGLM3-6B	100K	57.77	58.30
Qwen2.5-7B	20K†	75.44	74.68
LLaMA-3-8B	20K†	58.15	58.38

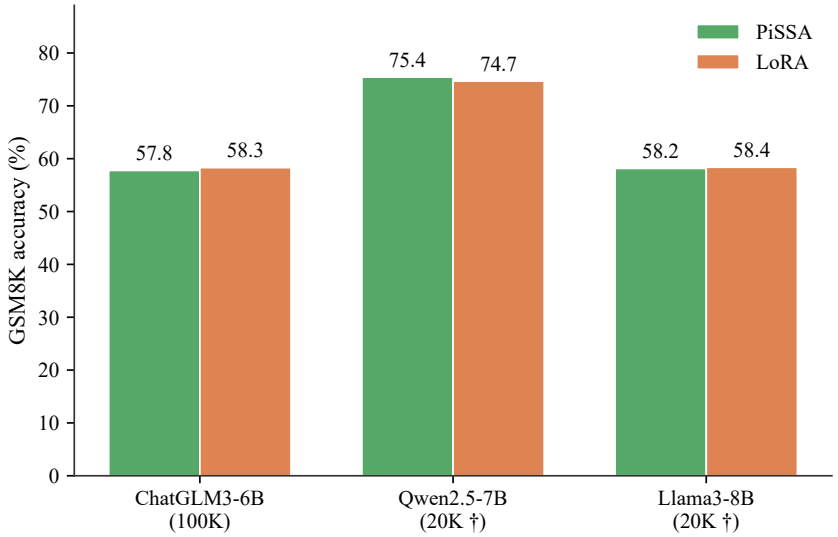


Figure 4: M2: PiSSA vs. LoRA across backbones. † denotes reduced 20K training and should be read as trend-only.

5.4. 4-bit quantization: QPiSSA vs. QLoRA (M3)

We study quantization along two axes. First, the *quantization-error reduction ratio*, defined exactly as in the original paper,

$$\text{reduction} \;=\; \Big(1 - \frac{\|W^{res} - \text{nf4}(W^{res})\|_*}{\|W - \text{nf4}(W)\|_*}\Big) \times 100\%,$$

where $\|\cdot\|_*$ is the nuclear norm, computed over all 224 linear layers and grouped by projection type (Table 3). QLoRA is 0% by definition. Our reproduction recovers the paper's *exact qualitative pattern*: the reduction is largest for the attention Q/K projections, smaller for V/O, and smallest for the MLP projections. The absolute magnitudes, however, are roughly half of those reported in the paper (average 10.71% vs. 19.4%); we analyze this in Section 6.

Table 3: M3: NF4 quantization-error reduction ratio (%) on LLaMA-2-7B, r=128, averaged over all layers by projection type. The per-projection pattern matches the paper exactly; absolute magnitudes are lower (see Section 6).

	Q	K	V	O	Gate	Up	Down	**AVG**
PiSSA (Meng et al., 2024)	27.9	27.2	18.7	18.6	15.8	13.6	13.6	**19.4**
Ours	18.97	18.10	8.88	8.91	8.22	5.94	5.96	**10.71**

Second, we run an end-to-end 4-bit training comparison on LLaMA-2-7B (Table 4, Figure 5). Here our result *diverges* from the paper's trend: QPiSSA reaches 22.97% GSM8K accuracy, below QLoRA's 27.22%, whereas the paper reports QPiSSA above QLoRA on LLaMA-3. We discuss the likely cause—our from-scratch NF4 dequantization path—in Section 6, and emphasize that the *mathematical* advantage of QPiSSA (Table 3) still holds.

Table 4: M3: end-to-end 4-bit training on LLaMA-2-7B (GSM8K). Our QPiSSA underperforms QLoRA, opposite to the paper's LLaMA-3 trend; see Section 6.

Method	GSM8K Acc. (%)	Final train loss
QLoRA	**27.22**	0.396
QPiSSA	22.97	0.411

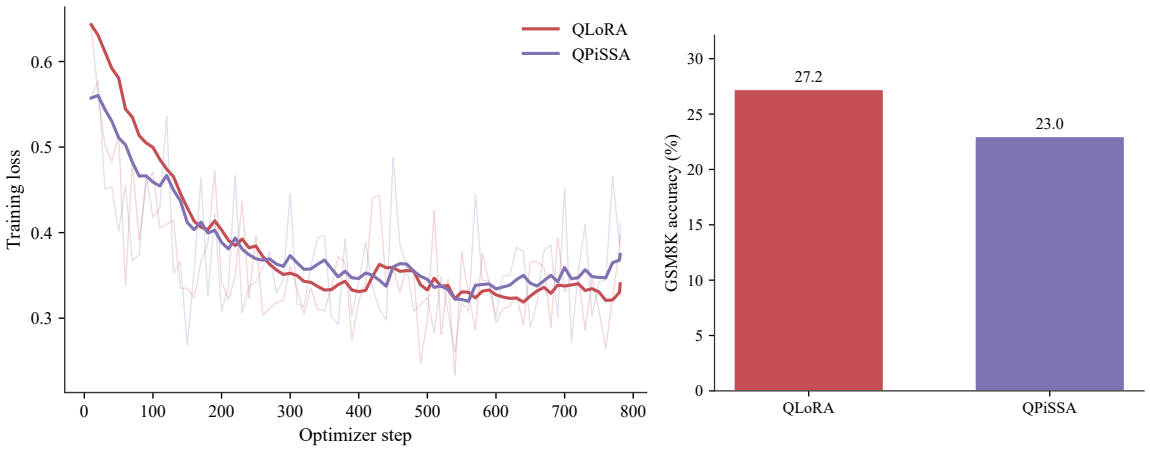


Figure 5: M3, LLaMA-2-7B 4-bit. Left: training-loss curves. Right: GSM8K accuracy for QLoRA and QPiSSA.

6. Discussion

What reproduces. The central claim of PiSSA reproduces cleanly. On the core LLaMA-2-7B benchmark, PiSSA equals full fine-tuning and beats LoRA by 7.5 points (Table 1); the toy and spectrum diagnostics behave exactly as described; the cross-architecture results show PiSSA is competitive with LoRA on diverse backbones; and the per-projection structure of the quantization-error reduction (Table 3) matches the paper layer-type for layer-type.

Comparison with the original numbers. Table 5 places our LLaMA-2-7B numbers next to the paper's Table 2. The *ordering* (PiSSA $\geq$ Full $>$ LoRA) is reproduced, but our absolute accuracies are uniformly lower (e.g. PiSSA 39.88% vs. 53.22%). Because the gap is consistent across all three methods, it most plausibly stems from evaluation-protocol differences (few-shot prompt format, generation length, answer-extraction regex), single-run vs. three-run averaging, and a 1024-token sequence cap, rather than from the adapter method itself. Under our compute budget we did not isolate each factor's contribution and leave this quantification to future work.

Table 5: Ours vs. the original paper on LLaMA-2-7B / GSM8K (MetaMathQA-100K, 1 epoch). The ordering reproduces; absolute values are lower for all methods (see text).

Method	PiSSA paper	Ours
Full FT	49.13	39.88
LoRA	43.23	32.37
PiSSA	53.22	39.88

Why our quantization numbers are smaller. The error-reduction magnitudes in Table 3 are about half the paper's, even though the pattern is identical. The most likely reasons are NF4 implementation details: our reduction is computed without double quantization (the paper's QLoRA setting enables it), and the set of layers averaged and the block size may differ from the original. We note that an earlier two-layer smoke test reported a much larger "79%" reduction; this turned out to be a sampling artifact, because those two layers were layer-0 Q/K, which have by far the highest reduction in the network. The full-network average is the honest number.

Why QPiSSA underperformed QLoRA in training. This is our clearest divergence from the paper. We attribute it to our from-scratch NF4 path: `NFPiSSALinear` dequantizes the residual on each forward pass through a CPU→GPU transfer and a Python-level dequantization, which is numerically weaker than the fused native CUDA 4-bit kernel that `bitsandbytes` uses for QLoRA. In other words, the *method* reduces quantization error as predicted (Table 3), but our *engineering* did not fully translate that into end-to-end accuracy. This is an implementation limitation, not evidence against PiSSA.

Limitations. (i) Qwen2.5-7B and LLaMA-3-8B were trained on only 20K samples and should be read as trends. (ii) Baichuan2-7B failed entirely: after training, both PiSSA and LoRA produced degenerate (zero-accuracy) generations. We traced this to architecture-specific incompatibilities (ALiBi, packed W_{pack} projections, and cache/RoPE/SDPA handling) that we could not resolve within the project's time and budget; we report it as a boundary of applicability under our pipeline rather than a flaw of PiSSA. (iii) Our absolute GSM8K accuracies and quantization magnitudes are lower than the paper's, as discussed above.

7. Conclusion

We independently reproduced PiSSA from scratch and verified its central claim: on LLaMA-2-7B, PiSSA matches full fine-tuning and clearly outperforms LoRA, and PiSSA decomposition provably reduces NF4 quantization error with the same per-projection structure reported in the original paper. We also reported, transparently, the places where our reproduction diverges—lower absolute GSM8K accuracy, smaller absolute quantization-error reduction, and a QPiSSA training result below QLoRA—and attributed each to evaluation-protocol, averaging, and from-scratch-implementation differences rather than to the method. Taken together, the reproduction supports PiSSA's core contribution while

delineating the practical conditions under which its quantitative numbers can be expected to hold.

Reproducibility. Our code (`src/pissa/`, `scripts/`), configuration files, fixed seeds, environment versions, and the exact training-subset sizes (100K / 20K) are released with this report, and all reported figures and tables are regenerated directly from the logged metrics under `results/`.

References

Fanxu Meng, Zhaohui Wang, and Muhan Zhang. PiSSA: Principal singular values and singular vectors adaptation of large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Yixiao Li, Yifan Yu, Chen Liang, Pengcheng He, Nikos Karampatziakis, Weizhu Chen, and Tuo Zhao. LoftQ: LoRA-fine-tuning-aware quantization for large language models. *arXiv preprint arXiv:2310.08659*, 2023.

Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James T. Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. MetaMath: Bootstrap your own mathematical questions for large language models. In *International Conference on Learning Representations (ICLR)*, 2024.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Hugo Touvron, Louis Martin, Kevin Stone, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.